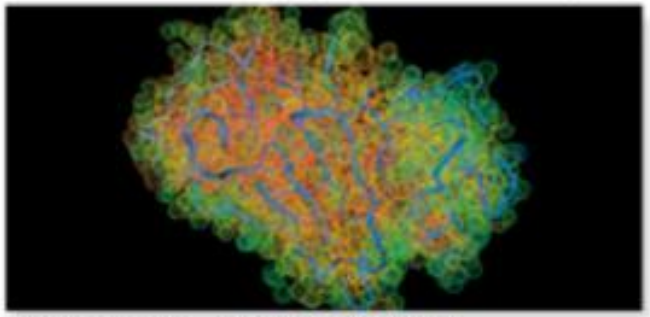
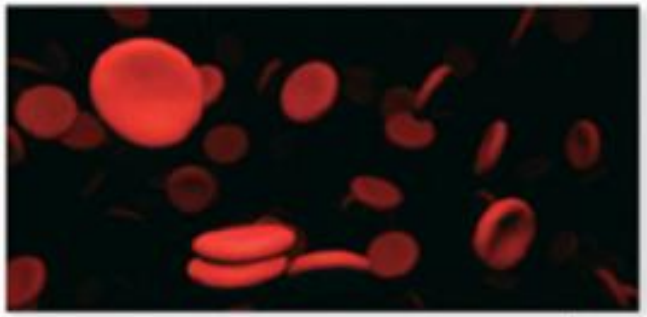
Proteins



Enzyme catalysis: space-filling model of an enzyme



Defense: venom



Transport: hemoglobin



Support: keratin



Motion: actin and myosin



Regulation: insulin



Storage: calcium

Proteins

-Are the most abundant organic molecules of the life.
-From *Greek Proteios: holding the first place*.
-Proteins are polymers of amino acids (Heteropolymers).
-Proteins generally contains more than 50 to several hundreds of amino acid units.

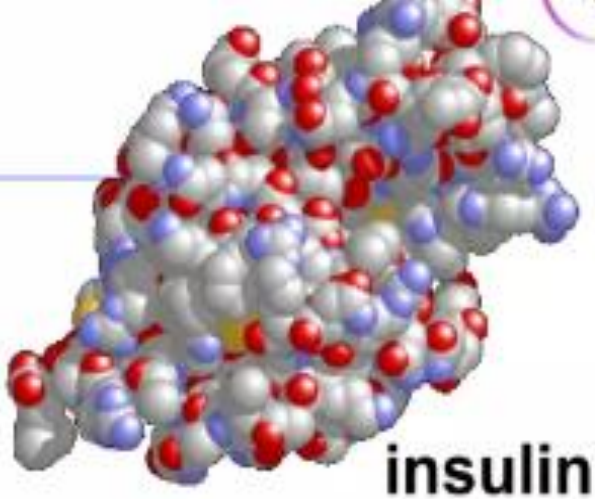
Elemental composition of Proteins:

Carbon- 50-55%
Hydrogen- 6-7.3%
Oxygen- 19-24%
Nitrogen- 13-19%
Sulphur- 0-4%
Also, contains
P, Fe, Cu, I, Mg,Mn, Zn



Proteins

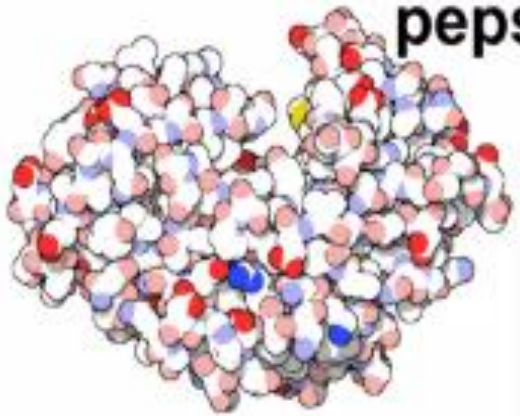




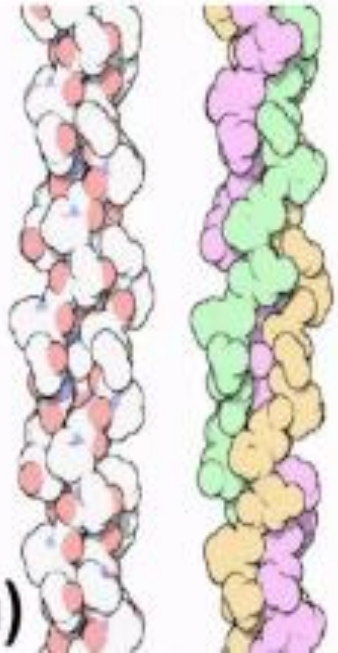
insulin

Examples

- muscle
- skin, hair, fingernails, claws
  - collagen, keratin
- pepsin
  - digestive enzyme in stomach
- insulin
  - hormone that controls blood sugar levels



pepsin



collagen (skin)

What are proteins and its components

- **Proteins are linear copolymers built from monomeric units called amino acids.**

- Twenty amino acids are commonly found in proteins. Among them 10 are **essential** (**body cannot synthesized them and must be obtained from the diet**).

- These amino acids contain a variety of different functional groups:
  - Alcohols (R-OH)
  - Phenols (Ph-OH)
  - Carboxylic acids (R-COOH)
  - Thiols (R-SH)
  - Amines ($R\text{-}NH_2$)
  - *and others…*

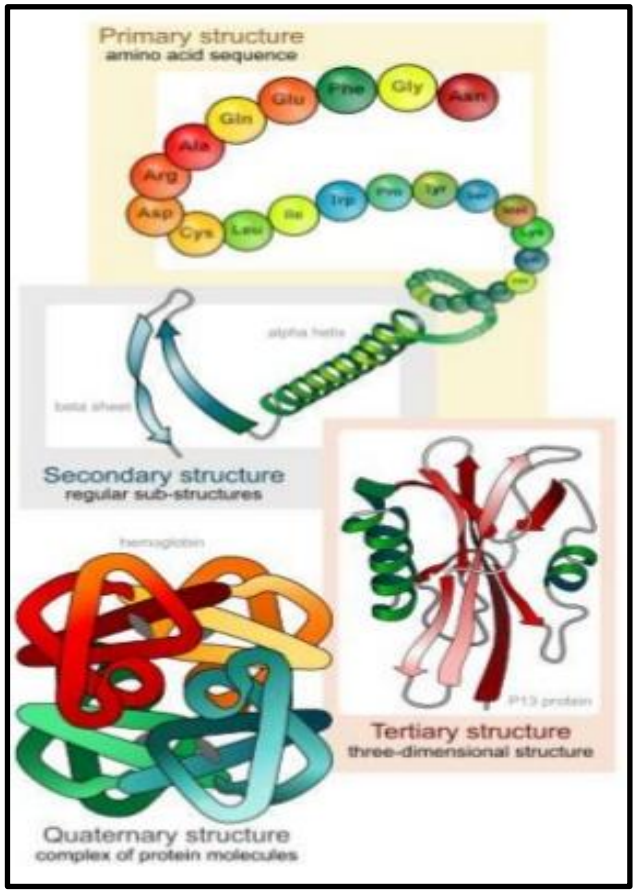
- **Protein function depends on both**
  - amino acid content, and
  - amino acid sequence.
- **Protein fold into diverse shapes to perform specific functions such as**
  - spherical
  - ellipsoidal
  - long strands, etc.
- **All information for 3-D structure is contained in the <u>linear sequence</u> of amino acids.**
- **There are four levels of structural organization of proteins:**

------Primary level (1° level)

------Secondary level (2° level)

------Tertiary level (3° level)

------Quaternary level (4° level)

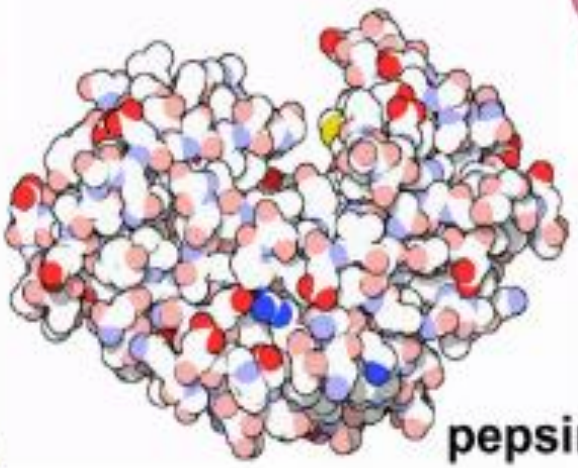


For proteins: SHAPE matters!

- **Proteins fold & twist into 3-D shape**
  - that's what happens in the cell!
- **Different shapes = different jobs**



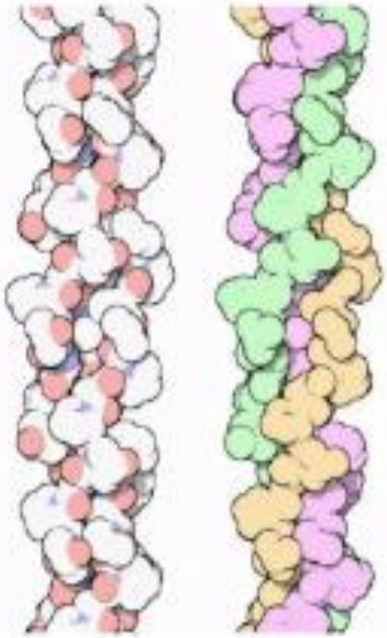
growth hormone



pepsin



hemoglobin



collagen

Biologically Important Proteins

Protein	No. of AA	Function
Insulin	51	Enzyme for sugar metabolism
Cytochrome C	104	Enzyme for cell respiration
Growth hormone	191	Used as anti-aging treatment
Hemoglobin	574	Oxygen transport in blood
Hexokinase	730	Enzyme for glycolysis
Gamma globulin	1320	Part of immune system in blood
Myosin	6100	Muscle action

Classification of protein
(Based on composition)

- **Simple protein**:*(depends on size and solubility)*
  - Fibrous protein (Ligament, Cartilage, hair, nail)
  - Globular protein
    - Albumin
    - Globulin
    - Histone
    - Gliadine

→ •The simplest
•Made of amino acid units only, joined by peptide bond
•Upon hydrolysis they yield mixture of amino acids and nothing else.

- **Conjugated protein:**
  - Nucleoprotein (RNA,DNA)
  - Liporpotein (Lipid)
  - Glycoprotein (CHO)
  - Rhodopsin (Retinol)
  - Feritin (Iron)
  - Hemoglobin (Hb heme)

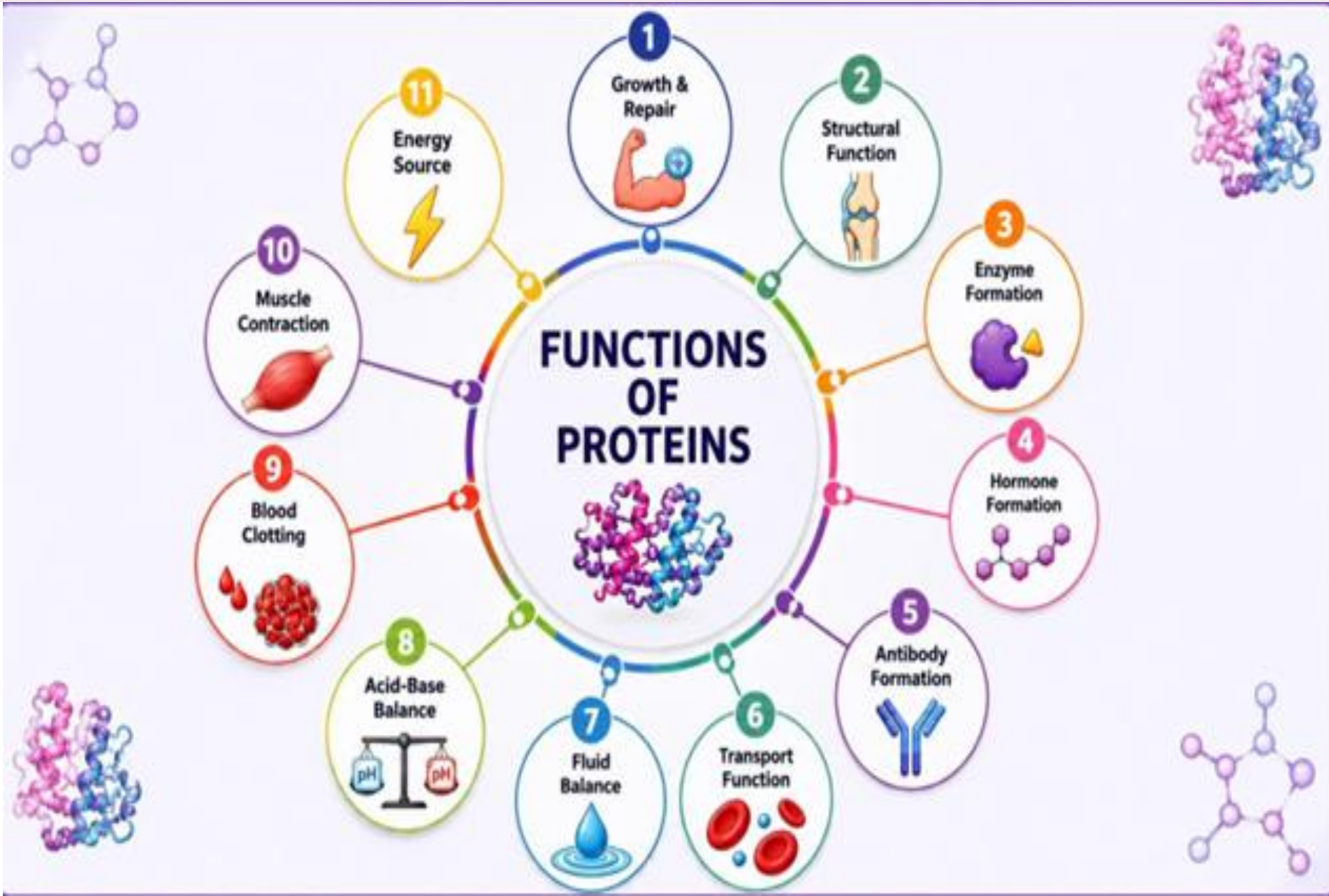
→ •Composed of simple proteins combined with a non-protein substance
•The non-proteinous substance is called **prosthetic group** or **cofactor**.

Derived proteins

•Peptones
•Peptides
•proteoses

→ •Not naturally occurring proteins
•Obtained from simple proteins by the action of enzymes and chemical agents.
•Results from hydrolysis of proteins

BIOLOGICAL FUNCTIONS OF PROTEINS



Building Blocks of Proteins
Amino Acids

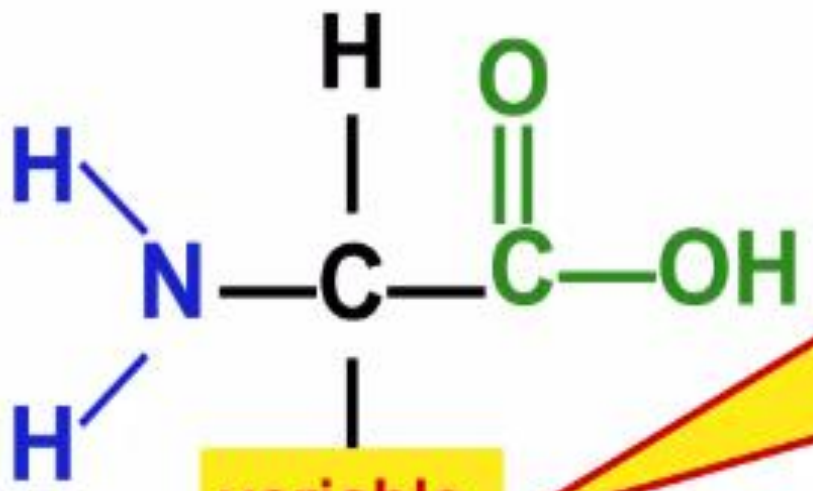
- Amino acids (monomers) are linked together to form proteins (polymers)
  - Each unique sequence of amino acids forms a different protein
  - All living things (even viruses) use the same 20 amino acids
- 20 different Amino Acids
  - Amino end (NH_2)
  - Carboxyl end (COOH)
  - Hydrogen
  - R group – variable component

Proteins

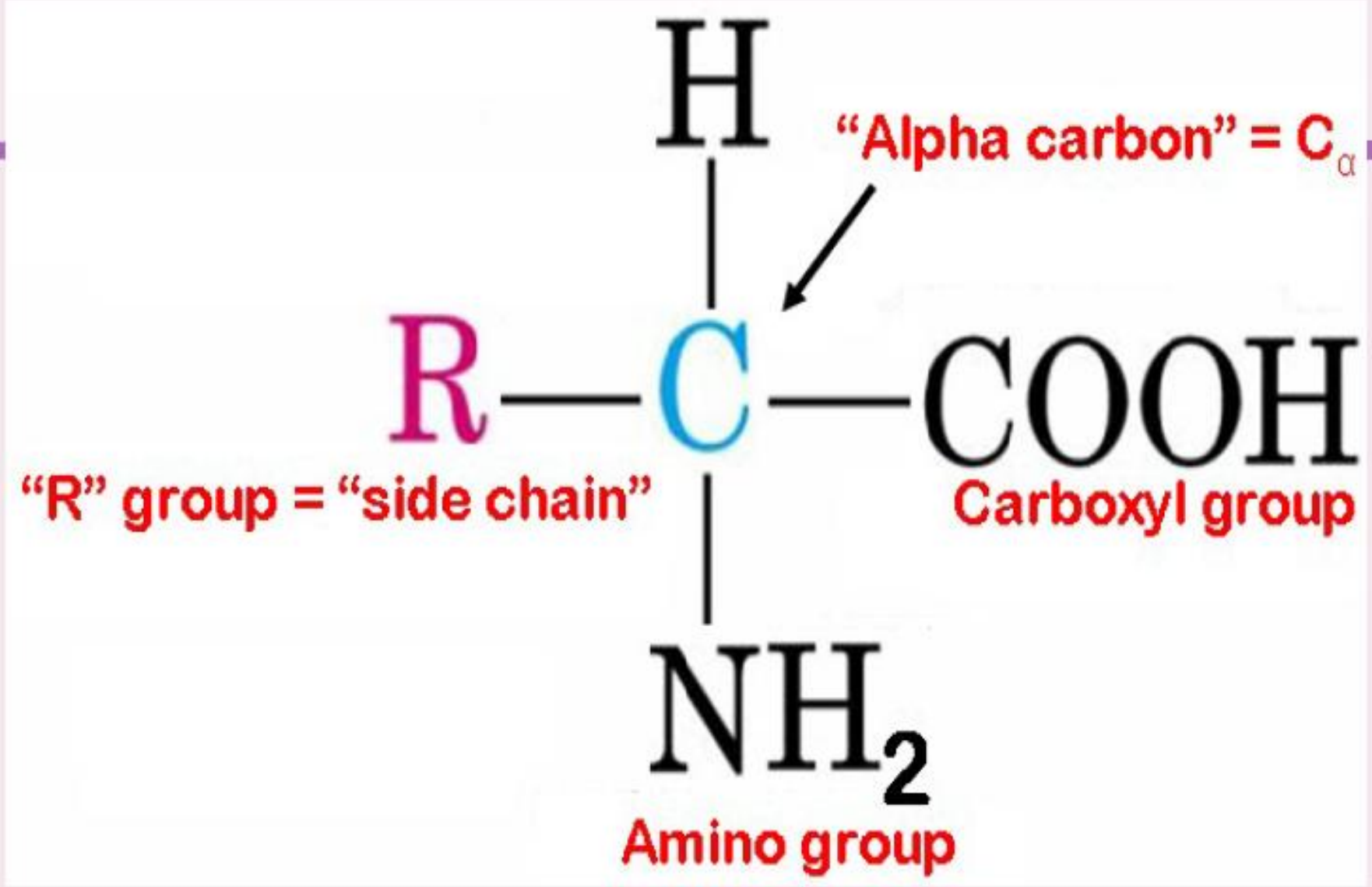
- Building block = amino acids

amino acid – amino acid – amino acid – amino acid – amino acid

- 20 different amino acids



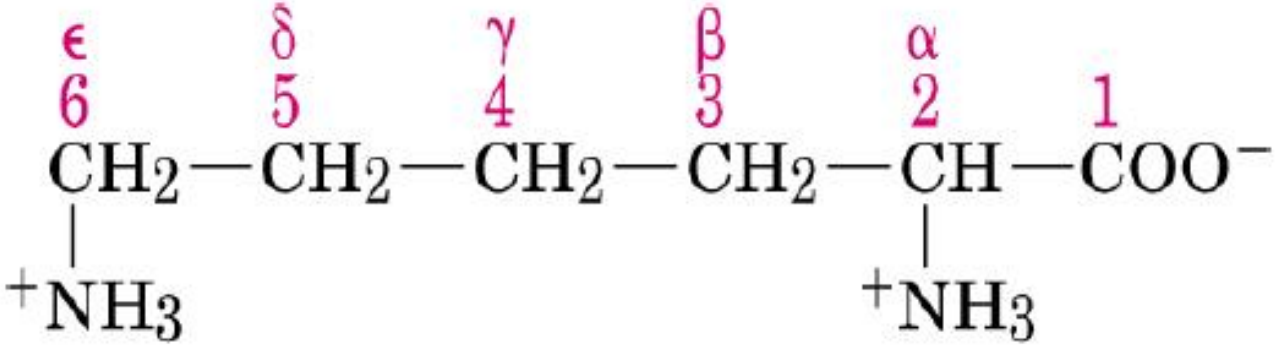
There's 20 of us... like 20 different letters in an alphabet! *Can make lots of different words*


H
"Alpha carbon" = Cα
R—C—COOH
"R" group = "side chain"
Carboxyl group
NH2
Amino group

Amino Acids: Atom Naming

- Organic nomenclature: start from one end
- Biochemical designation:
  - start from α-carbon and go down the R-group

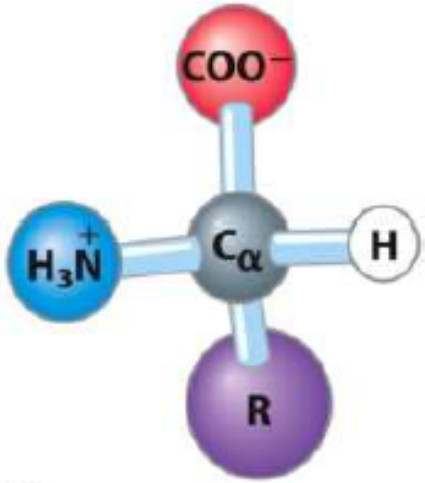
Lysine : Basic Amino Acid



Lysine

Most α–Amino Acids are Chiral

- The α-carbon has always four different substituents and is tetrahedral therefore exhibits optical isomerism
- Each amino acid has an unique fourth R-substituent
- Except In glycine, the R-substituent is also a hydrogen so it is non-chiral



Proteins only contain L amino acids

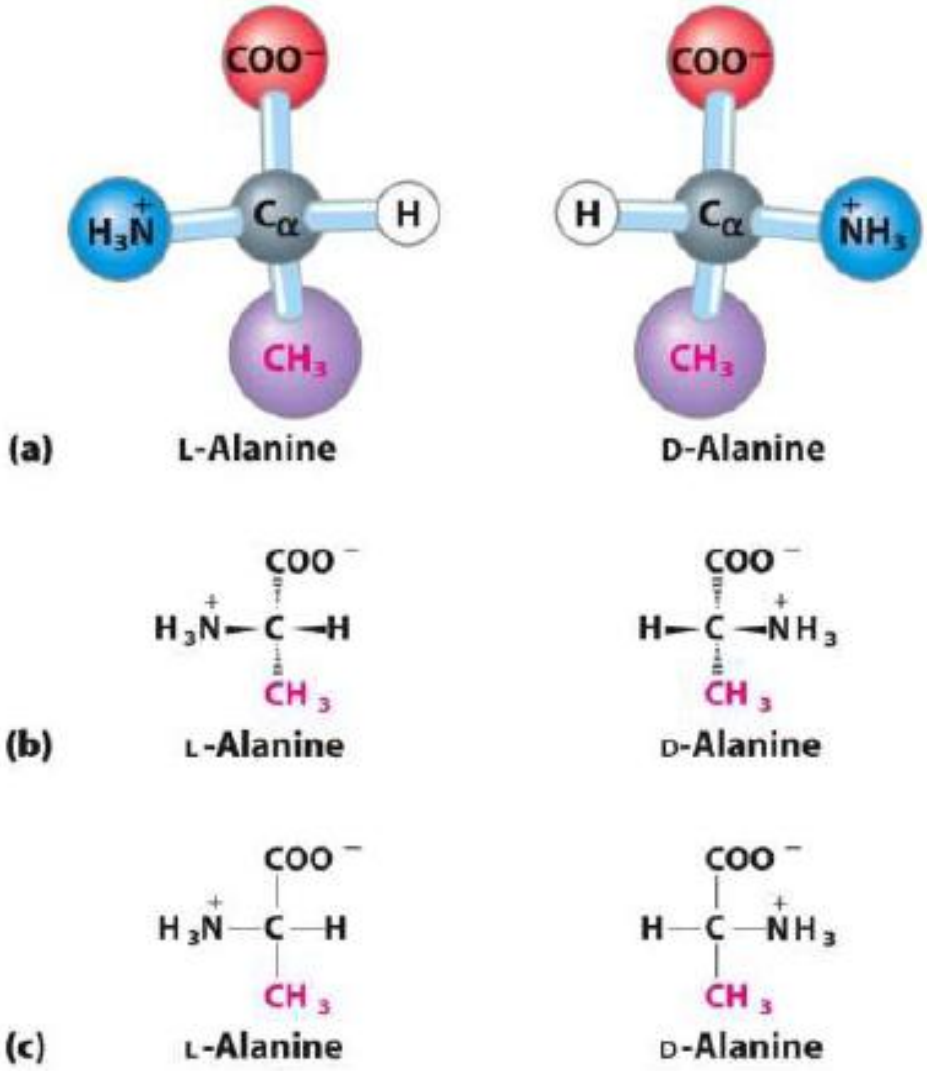


Figure 3-3
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

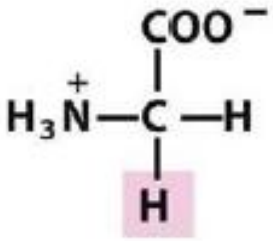
Amino Acids: Classification

A. Amino acids classification based on their structure

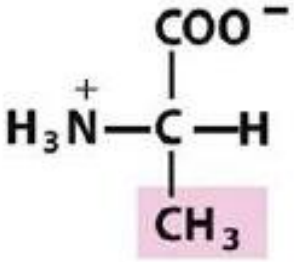
Common amino acids can be placed in seven basic groups depending on their R substituent

1. Aliphatic amino acids

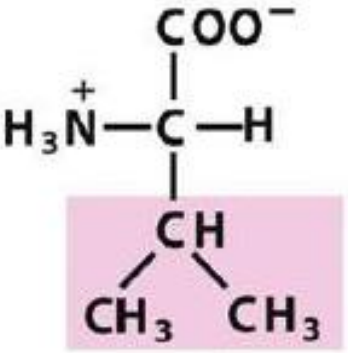
- Contains aliphatic R- chain
- these are hydrophobic and non-polar in nature
- these shows presence of simple or branched chain in R-group



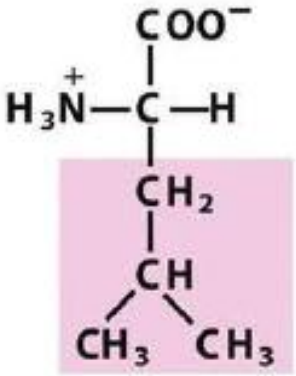
Glycine



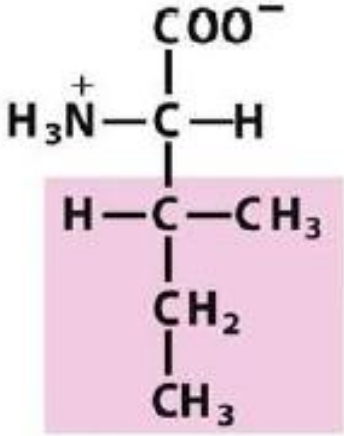
Alanine



Valine

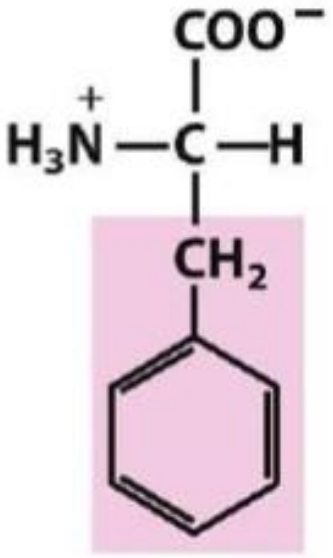


Leucine

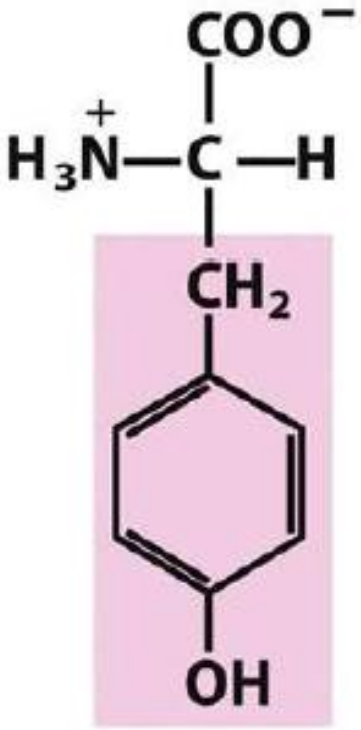


Isoleucine

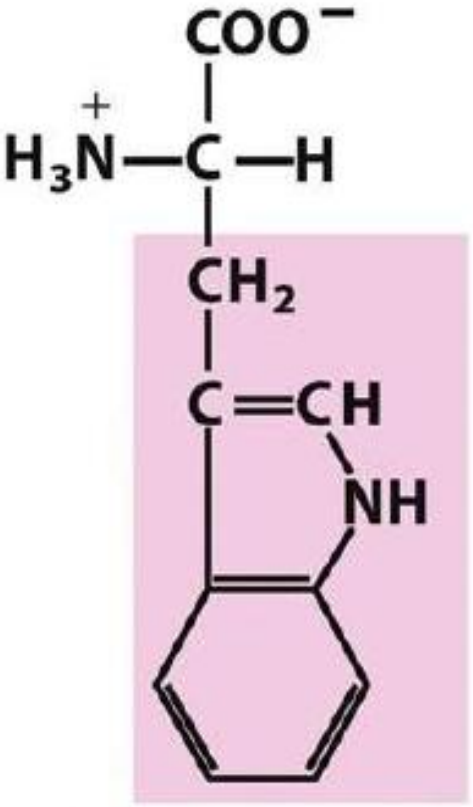
2. Aromatic amino acids
–Contains aromatic R– chain



Phenylalanine

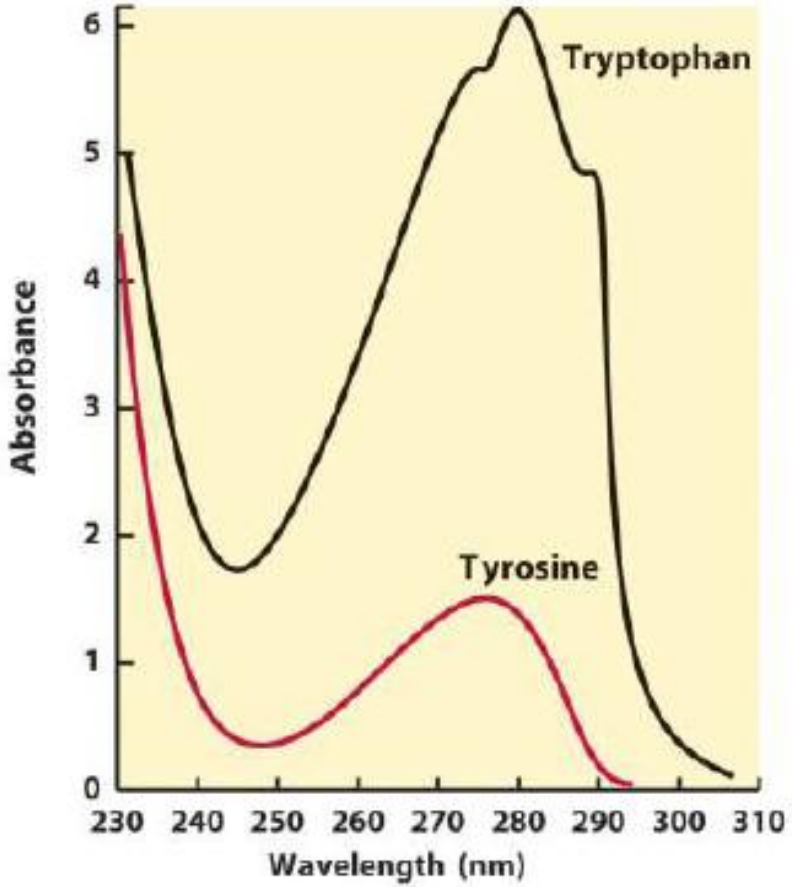


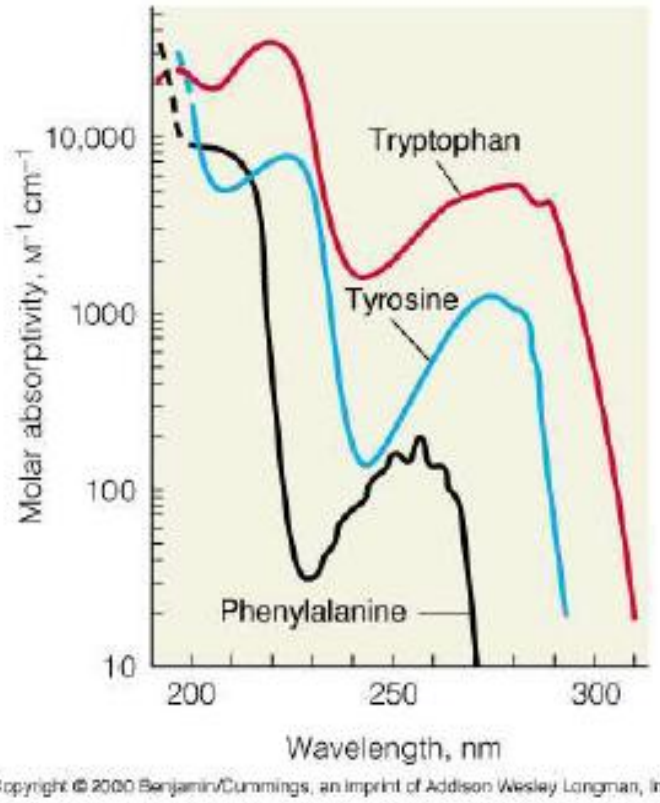
Tyrosine



Tryptophan

Absorption of UV Light by Aromatic Amino Acids

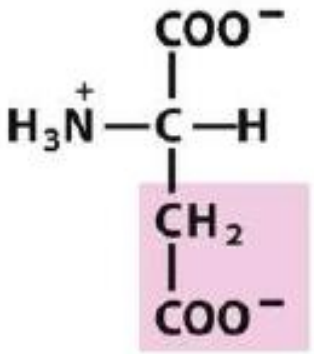




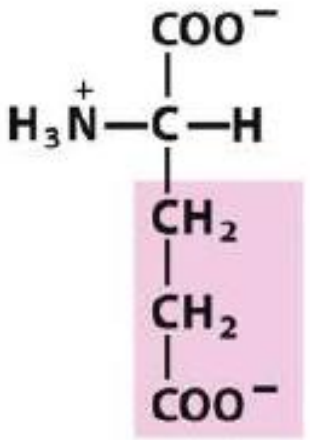
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3. Acidic amino acids

i) Dicarboxylic monoamino acids– Aspartic acid and Glutamic acid

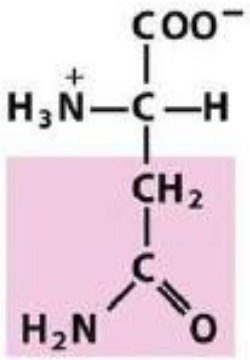


Aspartate

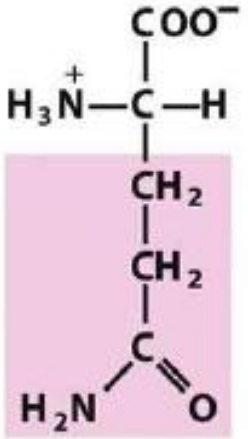


Glutamate

ii) Amides form of above acids



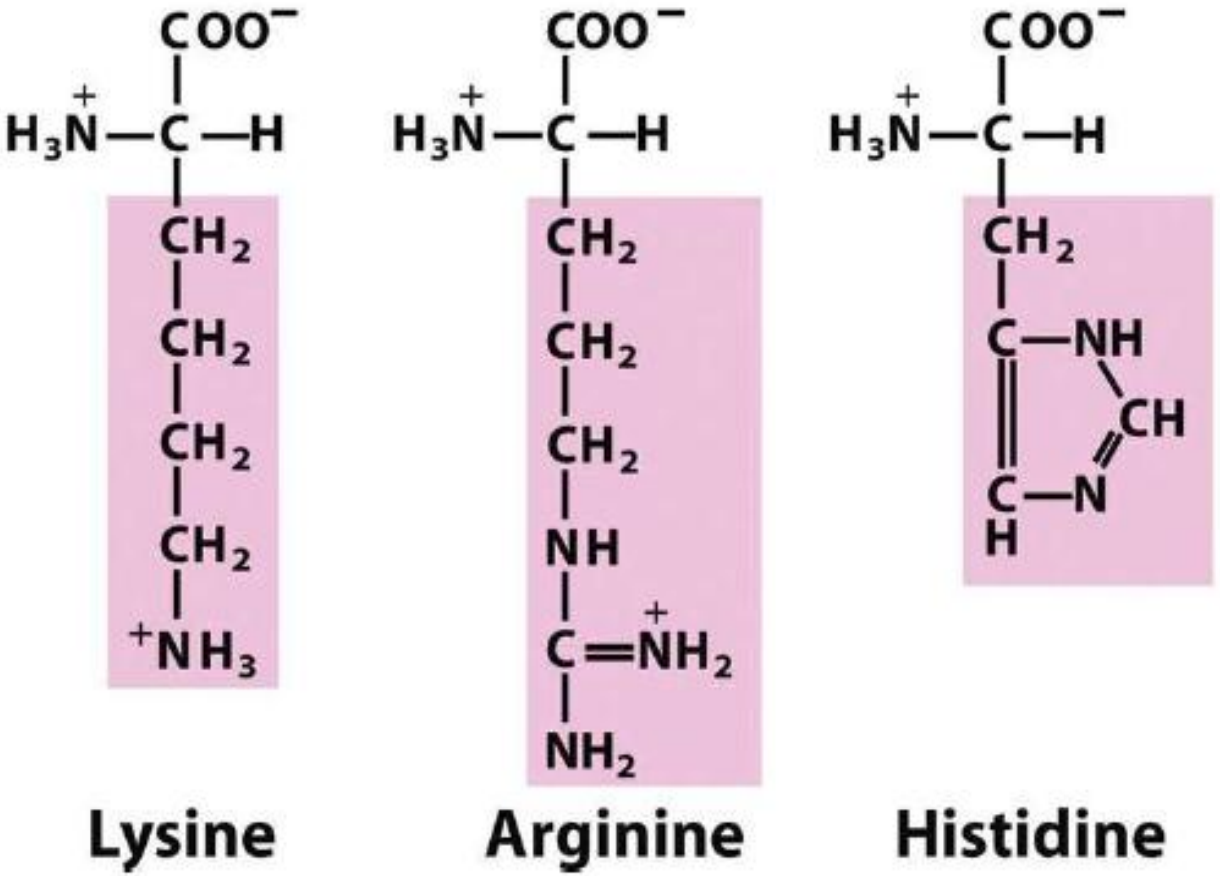
Asparagine



Glutamine

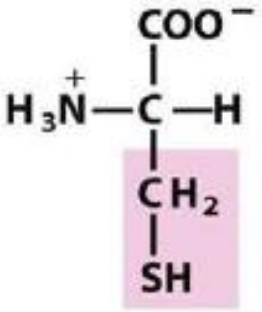
4. Basic amino acids

– These amino acids are basic in nature

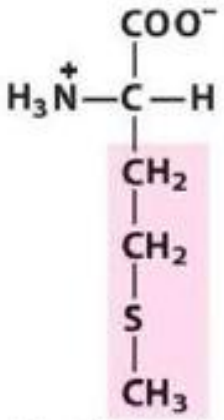


5. Sulphur containing amino acids

– These amino acids contain sulphur atom in the structure



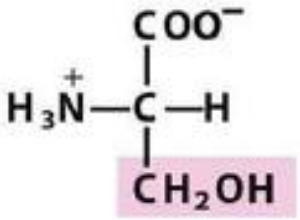
Cysteine



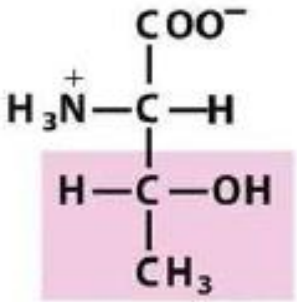
Methionine

6. Hydroxyl group containing amino acids

– These amino acids contain hydroxyl group in the structure



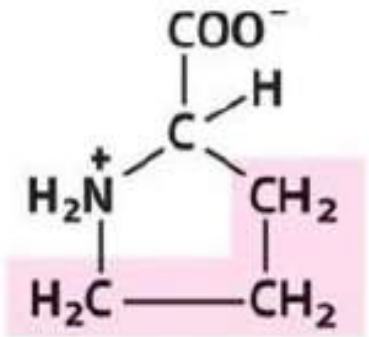
Serine



Threonine

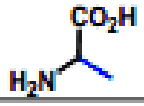
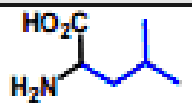
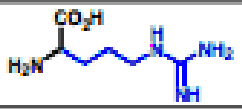
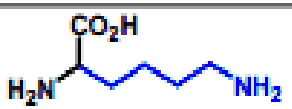
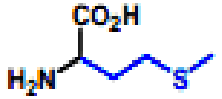
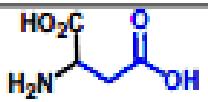
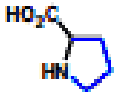
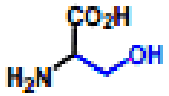
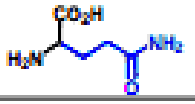
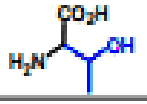
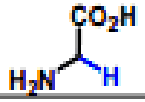
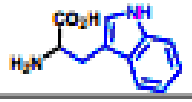
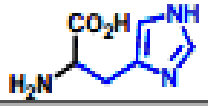
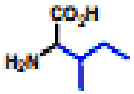
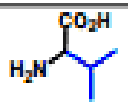
7. Imino acids

– These amino acids contain imino group(=NH) in place of amino group (–NH2) i.e. α– amino nitrogen is part of the structure



Proline

Standard Amino Acid Abbreviations

Name	Abbrev.	Structure*	**	Name	Abbrev.	Structure*	**
Alanine	Ala, A		H	Leucine	Leu, L		H
Arginine	Arg, R		B	Lysine	Lys, K		B
Asparagine	Asn, N		P	Methionine	Met, M		H
Aspartic Acid	Asp, D		A	Phenylalanine	Phe, F		H
Cysteine	Cys, C		H	Proline	Pro, P		H
Glutamic Acid	Glu, E		A	Serine	Ser, S		P
Glutamine	Gln, Q		P	Threonine	Thr, T		P
Glycine	Gly, G		H	Tryptophan	Trp, W		H
Histidine	His, H		B	Tyrosine	Tyr, Y		P
Isoleucine	Ile, I		H	Valine	Val, V		H

*Side Chain shown in Blue.

**Letters denote side chain properties: H = Hydrophobic, P = Polar, A = Acidic, B = Basic.

B. Amino acids classification based on their Nutritional requirements

1. **Essential amino acids**
   –the amino acids which can not be synthesized in the body but are required for normal functioning of body are called as essential amino acids.
   –These should be supplied through diet and supplements.
   –They required for proper growth and maintenance of the individual

-Valine	-Lysine
- Arginine	- Methionine
-Histidine	- Phenylalanine
- Isoleucine	- Threonine
- Leucine	- Tryptophan

2. **Non-essential amino acids**
   – these are synthesized in the body hence they need not be consumed in the diet

– Glycine	–Alanine
– Serine	– Cysteine
– Aspartic acid	– Glutamic acid
– Glutamine	– tyrosine
– Proline	– Asparagine

Some physical properties of amino acids

1. **Solubility**

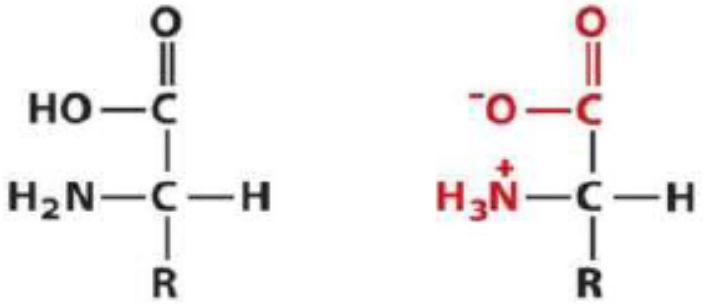
all amino acids are soluble in water, but their solubility varies to a great extent. Solubility depends on the nature of the R– group i.e. polarity of amino acids. Polar amino acids are highly soluble in water while non–polar amino acids are highly soluble in organic solvents like chloroform, ether, etc.

2. **Optical activity**

all standard amino acids except glycine have an asymmetric or chiral carbon atom. Due to this, amino acids are optically active. Also, they exist as stereo– isomers. All naturally occuring amino acids found in proteins are L– amino acids. Certain D– amino acids are found in some biological systems.

3. **Acid– base behavior**

amino acids contain the acidic carboxyl group (–COOOH) and the basic group amino (–NH2). Hence, amino acids are called as amphoteric molecules or ampholytes (i.e. amphoteric electrolyte).



Nonionic form **Zwitterionic form**

Zwitterion/dipolar ionic form- Can act as acid (proton donor) and base (proton acceptor)

Amphoteric (ampholytes)- non-ionic form

Functions or Role of amino acids

Amino acids are required for the body for various reasons:

1) For synthesis of various enzymes, hormones, plasma proteins and immuno-globulins

2) For the growth and repairs of body tissues

3) Source of energy when body is having inadequate supply of carbohydrates or fats

Zwitter ion or dipolar ion :

- **Zwitter ion is a hybrid molecule containing positive and negative ionic groups.**
- **The amino acids rarely exist in a neutral form with free carboxylic and free amino groups.**
- **In strongly acidic pH, the amino acid is positively charged (cation)**
- **In strongly alkaline pH, the amino acid is negatively charged (anion)**

Zwitterions

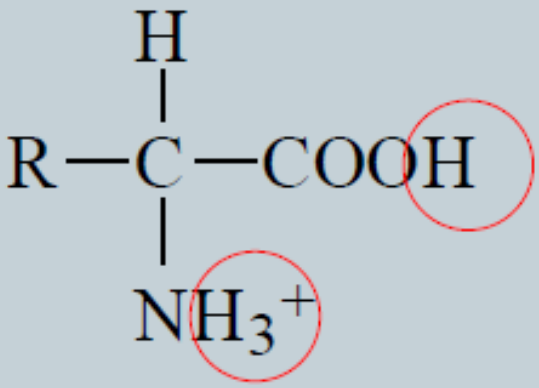
pH = 1-5

more acidic

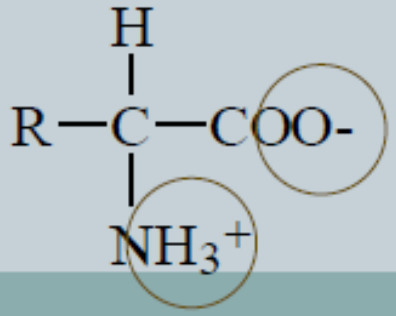
pH = 10-14

more basic

excess H^+



at pI (isoelectric point) charge = 0



excess OH^-

